

Does the LD panel matter for PGS?

Frederick J. Boehm
South Dakota State University

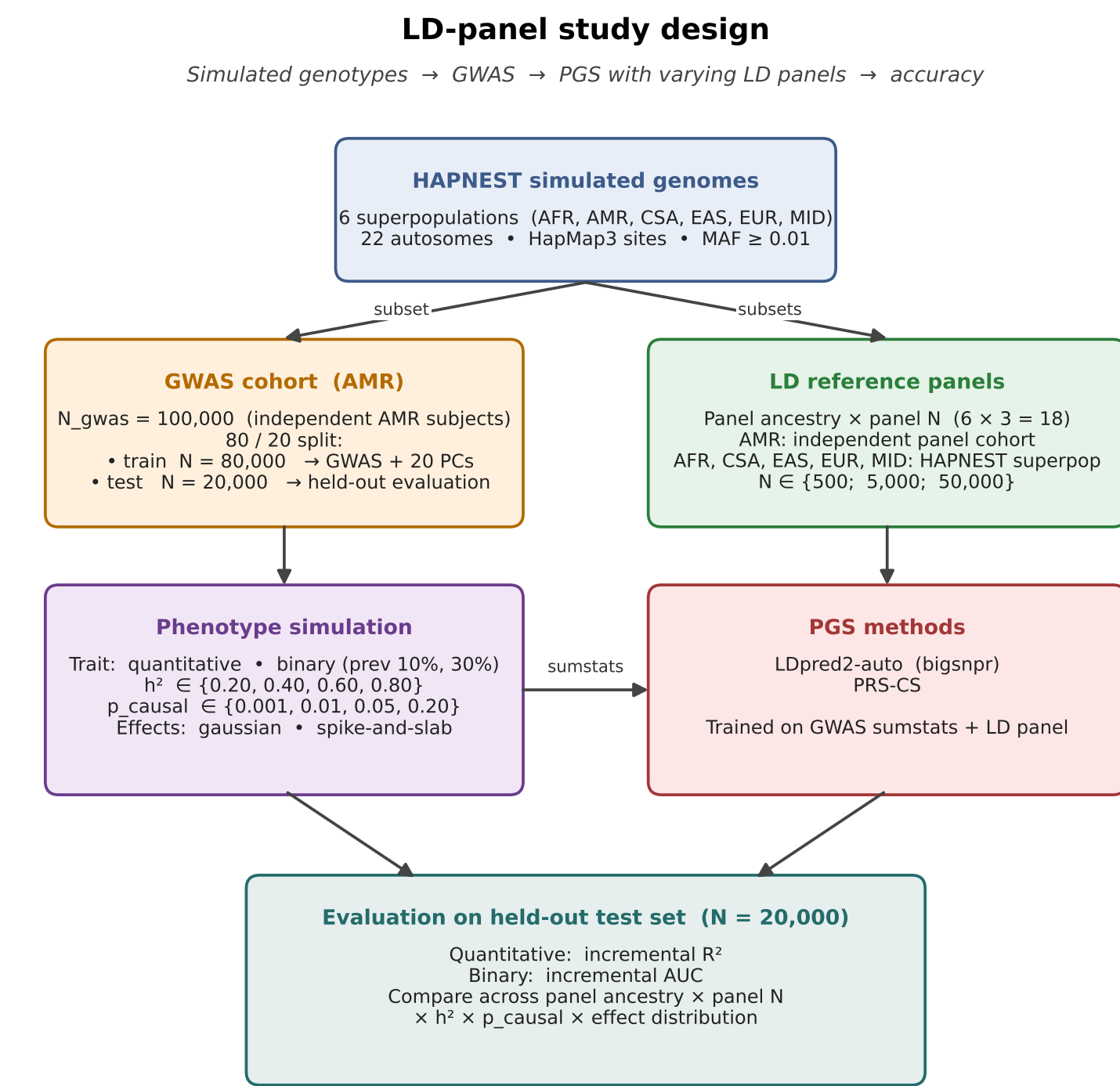
frederick.boehm@sdsstate.edu

github.com/fboehm/linkage-disequilibrium-panels

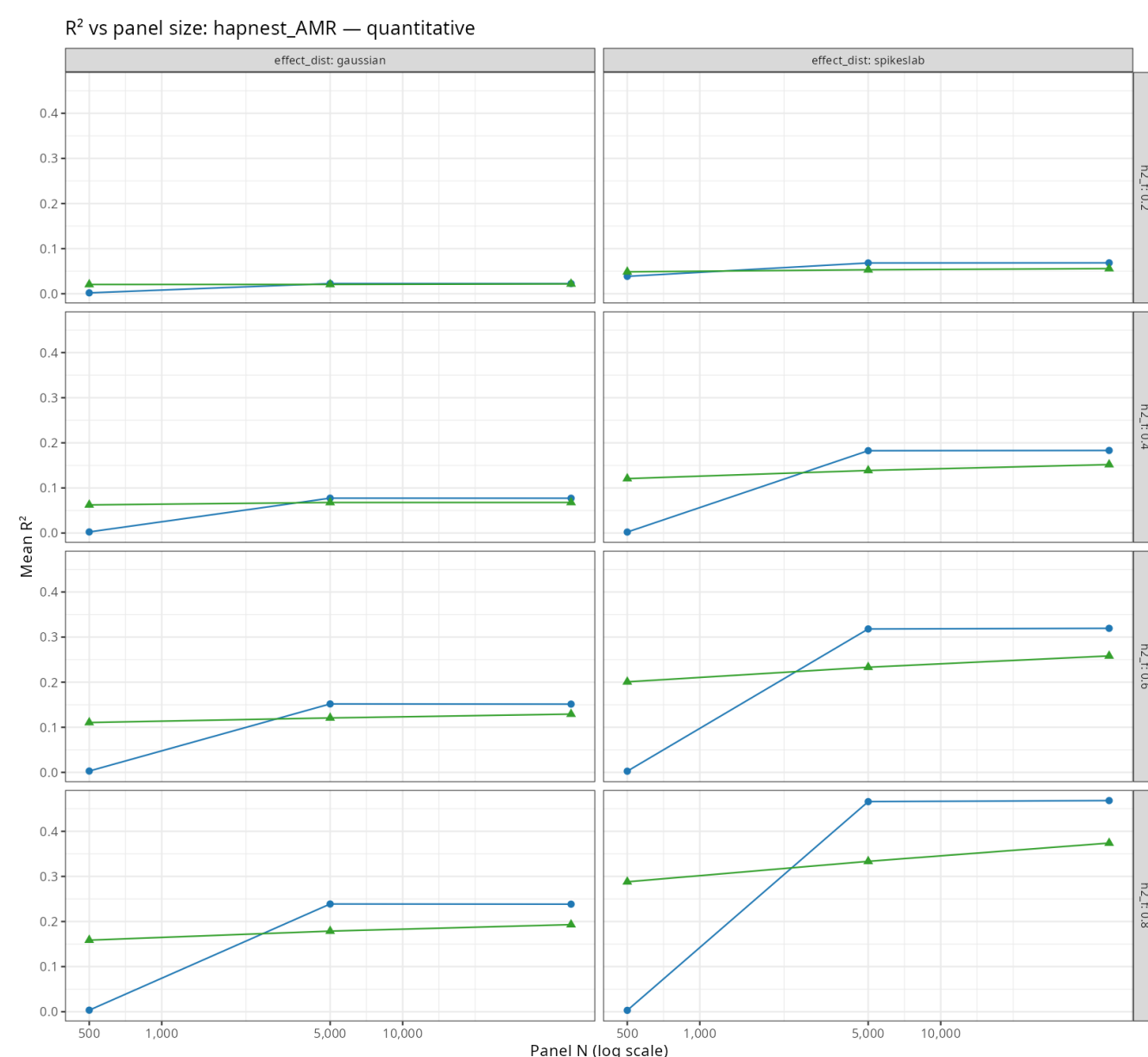
Background

Polygenic scores (PGS) trained on GWAS summary statistics need an external linkage-disequilibrium (LD) reference panel. In practice that panel is often **ancestry-mismatched** or **under-sized** relative to the GWAS. How much PGS accuracy do we lose?

Simulation design



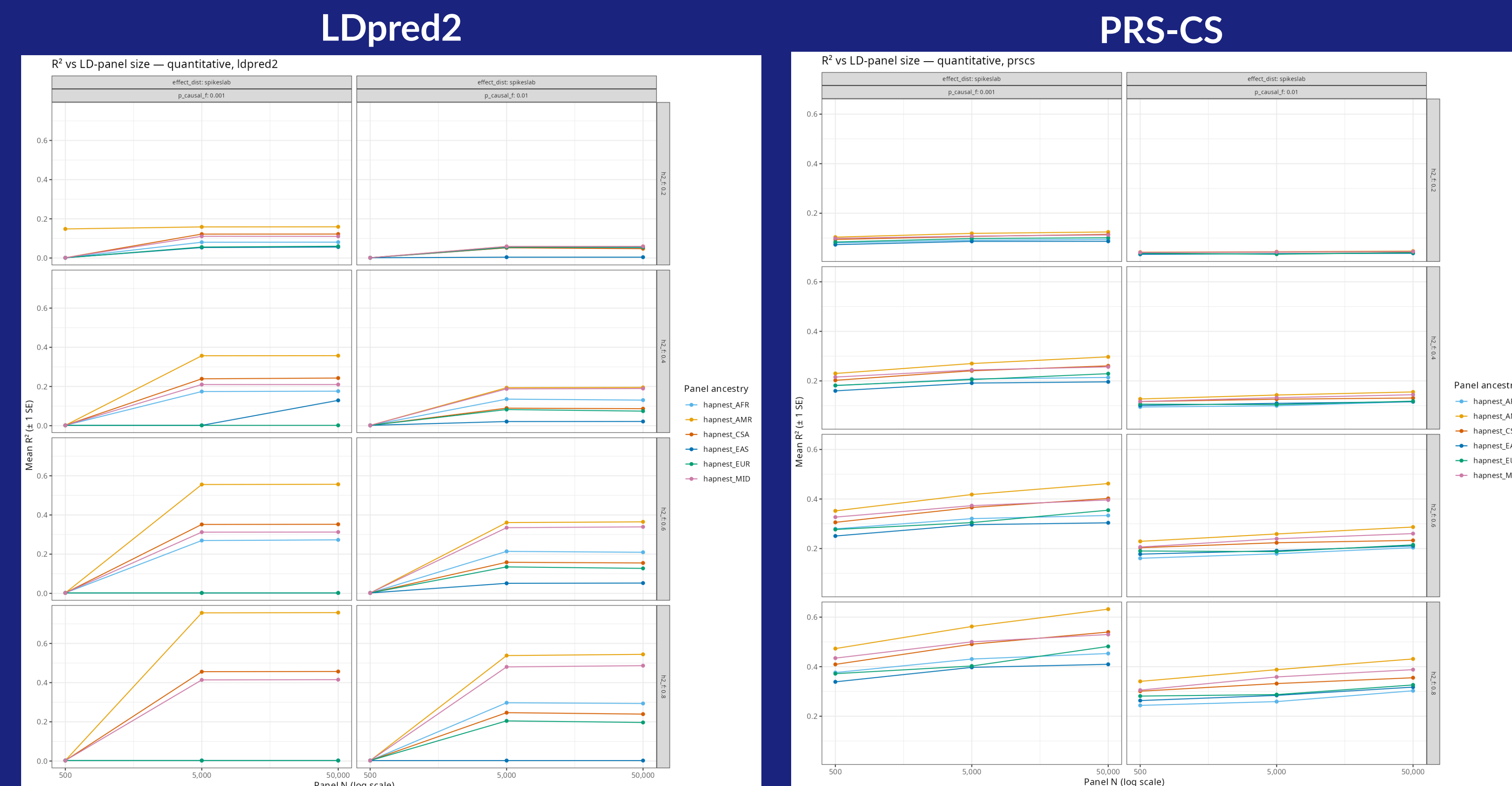
Methods compared



LDpred2 and **PRS-CS** produce similar results.

Panel ancestry and size matter
when the trait is **sparse** and **highly**
heritable.

In infinitesimal regimes, any panel will do.



R^2 vs. LD-panel size,
quantitative trait, sparse
spike-and-slab effects
($p_{\text{causal}} \in \{0.001, 0.01\}$).
Rows: heritability h^2
Matched AMR (orange)
beats mismatched EAS
(blue) by $\sim 2\times$ at high
 h^2 ; gap closes as panels
grow.

When the panel matters

Sparse, high- h^2 traits ($p_{\text{causal}} \leq 0.01$, $h^2 \geq 0.2$): matched-ancestry AMR panels reach $\sim 2\times$ the R^2 of the worst mismatched panel (EAS).

Panel size drives most of its gain between $N=500$ and $N=5,000$; the jump to $N=50,000$ adds little.

When it doesn't matter

Infinitesimal traits (gaussian effects): all six panel ancestries cluster within noise; panel size has negligible effect.

Low- h^2 traits ($h^2 \leq 0.10$): PGS accuracy is floor-limited; panel choice is a second-order concern.

Implications

For most polygenic traits the LD-panel choice is a *minor* contributor to PGS accuracy. Investment in matched-ancestry panels pays off **primarily for sparse, high-heritability traits**.

References

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GitHub repository



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